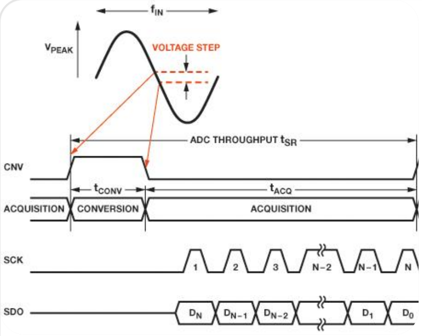
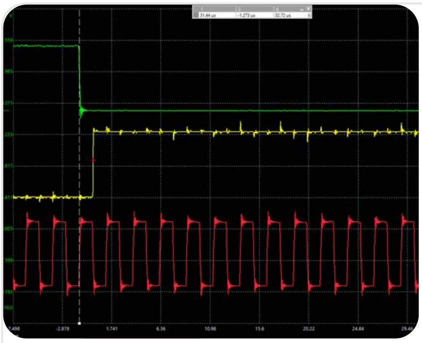
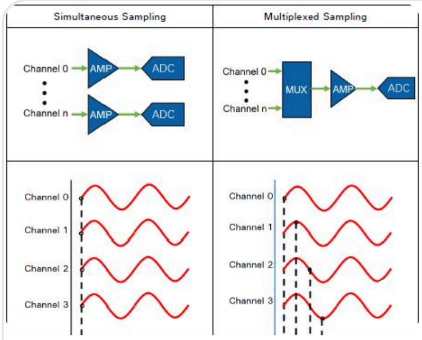
Great — you've shared the **exact timing diagrams** of the ADS8588H.
I'll explain **each figure step-by-step in a simple + exam-ready way** 👇

🔷 🔹 FIGURE 1 — CONVST Control Timing (Starting Conversion)







🧭 What is happening:

1. CONVSTA / CONVSTB ↓ (falling edge)

- This **starts conversion**
- Also performs **sampling (track → hold)**

👉 Important:

- Both signals allow **simultaneous sampling of channels**

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2. BUSY goes HIGH

- ADC is **working internally**
- SAR conversion happening

👉 This period = **tCONV (conversion time)**

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3. BUSY goes LOW

- Conversion **finished**
- Data is **ready to read**

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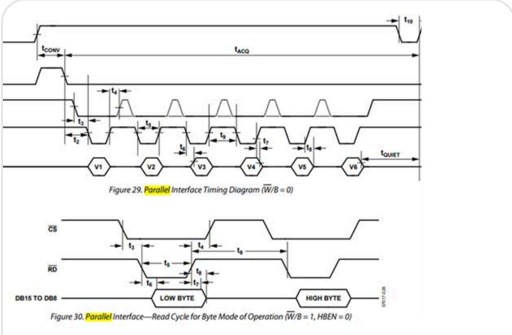
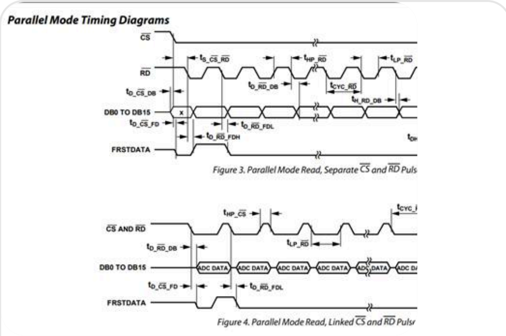
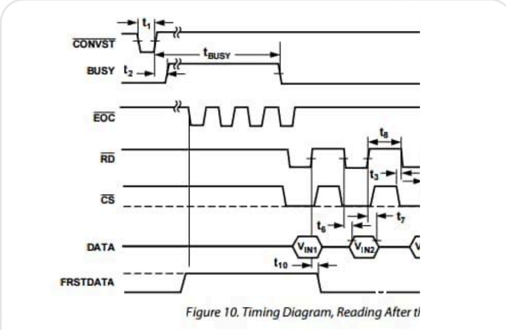
4. Timing parameters

- **tCYC** → total cycle time
- **tCONV** → conversion duration
- **tSU_BSYCS** → delay before safe read

🔷 🔹 FIGURE 2 — Data Read Operation







🧭 Two important cases:

✅ Case 1: Read during conversion

- **BUSY = HIGH**
- If you try reading → ❌ **invalid data**

👉 Marked as: *Read During Conversion*

✅ Case 2: Read after conversion

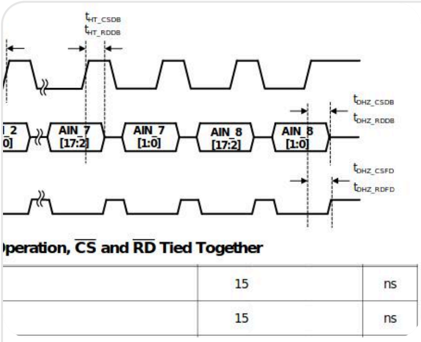
- Wait until **BUSY = LOW**
- Then:
  - **CS = LOW**
  - Data becomes valid

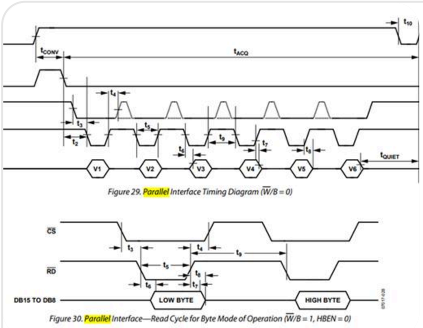
👉 This is the **correct method**

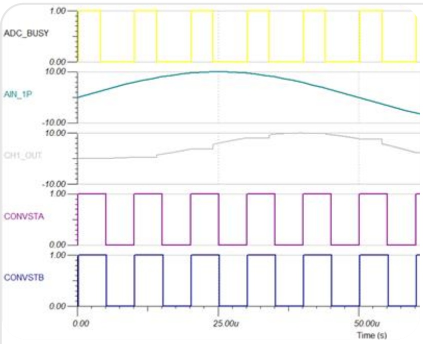
⚠️ Important timing:

- **tSU_BSYCS** → wait time after BUSY LOW before CS
- **tDZ_CNSC / tOZ_CSBY** → bus enable/disable delays

🔷 🔹 FIGURE 3 — Parallel Read (CS and RD Tied Together)







🧭 What is happening:

🔹 1. CS = RD (same signal)

- One pulse = both **select** + **read**

🔹 2. Each LOW pulse → new data

Every pulse gives:

Pulse	Output
1st	AIN_1 data
2nd	AIN_2 data
3rd	AIN_3 data
...	...
8th	AIN_8 data

👉 So **8 pulses = 8 channels**

🔹 3. DB[15:0]

- 16-bit data bus
- Outputs full data **in parallel**

🔹 4. FRSTDATA signal

- Indicates **first channel (AIN_1)**
- Helps synchronize reading

🔹 5. Timing parameters

- **tPL_RD / tPH_RD** → pulse width
- **tD_RDDB** → delay before data valid
- **tHZ_RDDB** → data bus goes high-Z

🔥 COMPLETE FLOW (VERY IMPORTANT)

```
CONVST ↓
   ↓
BUSY = HIGH (conversion running)
   ↓
BUSY = LOW (data ready)
   ↓
```

```
Apply CS/RD pulses
   ↓
AIN1 → AIN2 → ... → AIN8 data output
```

🎯 KEY UNDERSTANDING (EXAM POINT)

✔️ You must remember:

- **CONVST → starts conversion**
- **BUSY HIGH → ADC busy**
- **BUSY LOW → data ready**
- **CS/RD → read data**
- **Multiple pulses → multiple channels**

💡 SIMPLE ANALOGY

👉 Think like:

- **CONVST** → take photo
- **BUSY** → processing image
- **CS/RD pulses** → viewing each photo one by one

✅ FINAL ONE-LINE ANSWER

👉
CONVST starts conversion, BUSY shows status, and CS/RD pulses sequentially output each channel's 16-bit data on the parallel bus.